SAMPLING BASICS

Parameter (μ, σ^2): fixed, unknown, describes population.

Statistic $(\bar{X}, S^2)$: random variable computed from sample.

$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$

σ given $\Rightarrow$ use Z. s given $\Rightarrow$ use t.

CLT & SAMPLING DISTRIBUTIONS

From Normal Pop (exact)

$X_i \sim N(\mu, \sigma^2)$ iid $\Rightarrow \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ **exactly.**

$E(\bar{X}) = \mu$, $\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

CLT (any distribution, $n \geq 30$)

$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$, $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$\sum X_i \overset{\text{approx}}{\sim} N(n\mu, n\sigma^2)$

CLT STEP-BY-STEP

QUIZ Q1: $\mu = 78.3$ **kg**, $\sigma = 5.4$ **kg**, $n = 81$.
(a) Find $E(\bar{X})$ and $\mathrm{SD}(\bar{X})$.
Step 1: $E(\bar{X}) = \mu = \mathbf{78.3}$ kg. (Always equals μ!)
Step 2: $\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{5.4}{\sqrt{81}} = \frac{5.4}{9} = \mathbf{0.6}$ kg.

(b) Find $P(77.1 < \bar{X} < 79.5)$.
Step 1: Since $n = 81 \geq 30$, by CLT: $\bar{X} \sim N(78.3, 0.6^2)$.
Step 2: Standardize both bounds:
$z_{\text{low}} = \frac{77.1-78.3}{0.6} = \frac{-1.2}{0.6} = \mathbf{-2.0}$
$z_{\text{high}} = \frac{79.5-78.3}{0.6} = \frac{1.2}{0.6} = \mathbf{2.0}$
Step 3: Look up Z-table:
$P(77.1 < \bar{X} < 79.5) = P(-2 < Z < 2)$
$= \Phi(2) - \Phi(-2) = 0.9772 - 0.0228$
$= \mathbf{0.9544}$

Ex: CLT for $\sum X_i$ (passenger weights).
$\mu = 75$, $\sigma = 10$, $n = 100$. Find $P(\sum X_i > 7700)$.
Step 1: $\sum X_i \sim N(n\mu, n\sigma^2) = N(7500, 10000)$.
$\mathrm{SD} = \sqrt{10000} = 100$.
Step 2: $Z = \frac{7700-7500}{100} = \frac{200}{100} = 2.0$
Step 3: $P(\sum X_i > 7700) = P(Z > 2) = 1 - 0.9772 = \mathbf{0.0228}$

Ex: CLT with given pdf.
$f(x) = 2(1-x)$, $0 \leq x \leq 1$. $n = 200$, $S = \sum X_i$.
Step 1: $E(X) = \int_0^1 x \cdot 2(1-x)\,dx = \frac{1}{3}$
$E(X^2) = \int_0^1 x^2 \cdot 2(1-x)\,dx = \frac{1}{6}$
$V(X) = \frac{1}{6} - \left(\frac{1}{3}\right)^2 = \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$
Step 2: By CLT: $S \sim N\left(200 \cdot \frac{1}{3}, 200 \cdot \frac{1}{18}\right) = N(66.67, 11.11)$
$\mathrm{SD} = \sqrt{11.11} = 3.333$
Step 3: $P(S > 70) = P\left(Z > \frac{70-66.67}{3.333}\right) = P(Z > 1.0) = \mathbf{0.1587}$

TWO-SAMPLE DIFFERENCE

$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$

Variances always ADD!

$\sigma_A = \sigma_B = 1$, $n_1 = n_2 = 32$, $|\bar{x}_A - \bar{x}_B| = 1$.
Step 1: Under H_0: $\bar{X}_A - \bar{X}_B \sim N\left(0, \frac{1}{32} + \frac{1}{32}\right) = N(0, \frac{1}{16})$
$\mathrm{SD} = \sqrt{1/16} = 1/4 = 0.25$
Step 2: $Z = \frac{1-0}{0.25} = 4.0$
Step 3: $p = 2P(Z > 4) < 0.0001$. **Reject H_0.**

UNBIASED ESTIMATORS

Def: $\hat{\Theta}$ is unbiased for θ if $E(\hat{\Theta}) = \theta$.

$E(\bar{X}) = \mu$ ✓ $E(S^2) = \sigma^2$ ✓

Efficient: smallest variance among all unbiased.

Std Error: $\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

QUIZ Q2: RS $X_1, \ldots, X_n$ with mean μ, var σ^2.
$\hat{\mu} = w_1(X_1 + X_2) + w_2 \sum_{i=3}^{n} X_i$
Find condition on w_1, w_2 so $\hat{\mu}$ is unbiased for μ.
Step 1: Take expectation:
$E(\hat{\mu}) = w_1(E(X_1) + E(X_2)) + w_2 \sum_{i=3}^{n} E(X_i)$
$= w_1(\mu + \mu) + w_2(n-2)\mu$
$= 2w_1\mu + (n-2)w_2\mu$
$= [2w_1 + (n-2)w_2]\,\mu$
Step 2: For unbiased: $E(\hat{\mu}) = \mu$, so we need:
$\boxed{2w_1 + (n-2)w_2 = 1}$
Any w_1, w_2 satisfying this are valid.
E.g., $w_1 = w_2 = \frac{1}{n}$ gives the usual $\bar{X}$.

HYPOTHESIS TESTING

5 Steps (always show all!)

1. State H_0, H_a 2. Test stat 3. Rejection region / p-value 4. Decision 5. Conclusion

Significance Level & Power

$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$

$\text{Power} = P(\text{reject } H_0 \mid H_1 \text{ true})$

Ex: $X_i \sim N(\mu, 25)$, $n = 100$. Test $H_0: \mu = 10$ vs $H_1: \mu \neq 10$.
Reject if $\bar{X} \geq 10.98$ or $\bar{X} \leq 9.02$.
Find α: Under H_0: $\bar{X} \sim N(10, 25/100) = N(10, 0.25)$.
$\mathrm{SD}(\bar{X}) = \sqrt{0.25} = 0.5$
$\alpha = P(\bar{X} \geq 10.98) + P(\bar{X} \leq 9.02)$
$= 2P\left(Z > \frac{10.98-10}{0.5}\right) = 2P(Z > 1.96)$
$= 2(0.025) = \mathbf{0.05}$

Find power at $\mu = 10.5$:
Under H_1: $\bar{X} \sim N(10.5, 0.25)$, $\mathrm{SD} = 0.5$.
$\text{Power} = P(\bar{X} \geq 10.98 \mid \mu = 10.5) + P(\bar{X} \leq 9.02 \mid \mu = 10.5)$
$= P\left(Z > \frac{10.98-10.5}{0.5}\right) + P\left(Z < \frac{9.02-10.5}{0.5}\right)$
$= P(Z > 0.96) + P(Z < -2.96)$
$= 0.1685 + 0.0015 = \mathbf{0.17}$

Binomial + Normal Approx

$X \sim \mathrm{Bin}(n, p)$: approx $N(np, npq)$

Continuity corr: $P(X \geq k) \approx P\left(Z \geq \frac{k-0.5-np}{\sqrt{npq}}\right)$

CI $\Leftrightarrow$ Hypothesis Test

μ_0 inside CI $\Leftrightarrow$ fail to reject H_0

μ_0 outside CI $\Leftrightarrow$ reject H_0

CONFIDENCE INTERVALS

CI for μ (σ known) — use Z

$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

Step-by-step: $\bar{x} = 5.027$, $\sigma = 0.1$, $n = 100$, 95% CI.
Step 1: Confidence $= 0.95$, so $\alpha = 0.05$, $\alpha/2 = 0.025$.
Step 2: $z_{0.025} = 1.96$ (from table).
Step 3: Margin of error: $E = 1.96 \cdot \frac{0.1}{\sqrt{100}} = 1.96 \cdot 0.01 = 0.0196$
Step 4: CI: $5.027 \pm 0.0196 = [\mathbf{5.007}, \mathbf{5.047}]$

CI for μ (σ unknown) — use t

$\bar{X} \pm t_{n-1,\,\alpha/2} \cdot \frac{S}{\sqrt{n}}$

Step-by-step: $n = 21$, $\bar{x} = 8.232$, $s = 0.026$, 95% CI.
Step 1: σ unknown $\Rightarrow$ use t. $\mathrm{df} = n - 1 = 20$.
Step 2: $t_{0.025,20} = 2.086$ (from t-table).
Step 3: $E = 2.086 \cdot \frac{0.026}{\sqrt{21}} = 2.086 \cdot 0.00567 = 0.0118$
Step 4: CI: $8.232 \pm 0.012 = [\mathbf{8.220}, \mathbf{8.244}]$

Two-Sample CI (pooled t)

$(\bar{X}_1 - \bar{X}_2) \pm t_{n_1+n_2-2,\,\alpha/2} \cdot S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$

Only when $\sigma_1^2 = \sigma_2^2$.

$n_1 = 8$, $n_2 = 7$, $\bar{x}_A = 7.70$, $\bar{x}_B = 8.04$, $s_A = 0.85$, $s_B = 0.458$. 95% CI.
Step 1: $s_p^2 = \frac{7(0.85)^2 + 6(0.458)^2}{8+7-2} = \frac{5.0575+1.2590}{13} = 0.486$
Step 2: $\mathrm{df} = 13$, $t_{0.025,13} = 2.16$
Step 3: $(7.70 - 8.04) \pm 2.16\sqrt{0.486(\frac{1}{8} + \frac{1}{7})}$
$= -0.34 \pm 2.16\sqrt{0.486 \cdot 0.268} = -0.34 \pm 0.779$
Step 4: $= (\mathbf{-1.119}, \mathbf{0.439})$. Contains 0: **fail to reject**.

Other CIs

Proportion: $\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Variance: $\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right)$

Sample size: $n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$

CHEBYSHEV & MARKOV

Markov ($X \geq 0$, only need $E[X]$): $P(X \geq a) \leq \frac{E(X)}{a}$

Chebyshev (any dist, need μ, σ^2): $P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$

$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$

$\mu = 5$, $\sigma^2 = 4$, $n = 100$. Find lower bound on $P(3 < \bar{X} < 7)$.
Step 1: $\mathrm{Var}(\bar{X}) = 4/100 = 0.04$, $\mathrm{SD}(\bar{X}) = 0.2$.
Step 2: Interval is $\mu \pm 2 = 5 \pm 2$. So $k = \frac{2}{0.2} = 10$.
Step 3: $P(3 < \bar{X} < 7) \geq 1 - \frac{1}{10^2} = \mathbf{0.99}$

WLLN Proof

$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \to 0$

$\therefore \bar{X}_n \to \mu$ in probability. $\square$

χ^2, t, F DISTRIBUTIONS

Chi-Squared (χ^2)

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$ (normal pop)

$E[\chi^2(\nu)] = \nu$, $\mathrm{Var} = 2\nu$

Additivity: $\chi^2(\nu_1) + \chi^2(\nu_2) \sim \chi^2(\nu_1 + \nu_2)$

$\bar{X}$ and S^2 are independent (normal pop)

$n = 10$, $\sigma^2 = 2$. Find $P(S^2 > 3.5)$.
$\frac{9S^2}{2} \sim \chi^2(9)$. Want $P\left(\chi^2(9) > \frac{9 \cdot 3.5}{2}\right) = P(\chi^2(9) > 15.75)$
Table: $\chi^2_{0.10}(9) = 14.68$, $\chi^2_{0.05}(9) = 16.92$.
$14.68 < 15.75 < 16.92$, so $\mathbf{0.05 < P < 0.10}$ (≈ 0.07).

t-Distribution

$T = \frac{Z}{\sqrt{V/\nu}}$, $Z \sim N(0,1)$, $V \sim \chi^2(\nu)$ indep.

One-sample: $T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$

$E(T) = 0$, $\mathrm{Var}(T) = \frac{\nu}{\nu-2}$. As $\nu \to \infty$: $t \to N(0,1)$.

Table: $P(T > t_\alpha) = \alpha$. Symmetric about 0.

F-Distribution

$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1, n_2-1)$

Under $H_0: \sigma_1^2 = \sigma_2^2$: $F = S_1^2/S_2^2$

Reciprocal: $f_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{f_\alpha(\nu_2, \nu_1)}$

WHEN TO USE WHAT

Situation	Stat	Dist
μ, σ known	$Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$	$N(0,1)$
μ, σ unknown	$T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$	$t(n-1)$
Test σ^2	$\frac{(n-1)S^2}{\sigma_0^2}$	$\chi^2(n-1)$
Compare σ^2's	$\frac{S_1^2}{S_2^2}$	$F(n_1-1, n_2-1)$
$n \geq 30$, any	either	$\approx N(0,1)$

Key z-values

90%: $z_{0.05} = 1.645$ 95%: $z_{0.025} = 1.96$

99%: $z_{0.005} = 2.576$

Common Traps

- CLT makes $\bar{X}$ normal, **NOT the raw data**
- μ is a fixed number, not a r.v.
- $E(\bar{X}) = \mu$ regardless of n
- $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$ *decreases* with n
- SD of $\hat{p} = \sqrt{p(1-p)/n}$, **not** $\sqrt{np(1-p)}$
- $n = 4$ from skewed pop $\neq$ normal approx
- Divide by $n-1$ (not n) for S^2 to be unbiased